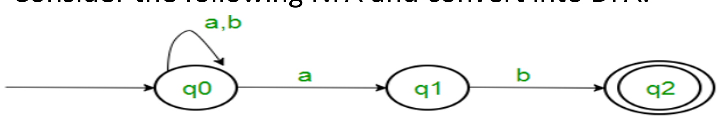
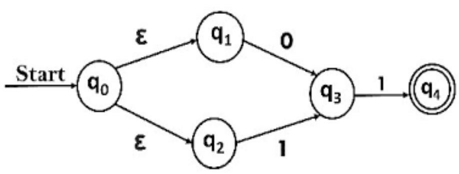
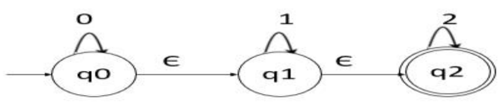
School of Engineering-A.Y 2025-2026 Regulation: R23

(CSE-COMPUTER SCIENCE & ENGINEERING)

III Year B. Tech - I Semester

Subject Name: Compiler Design Subject Code: MR23-1CS0116

QUESTION BANK

Qno	Question	Marks	Section
1	a) Define Finite Automata and explain its representations with example. b) Design a FA which i) starts with 1 and ends with 0 ii) contains 2 consecutive a's and 2 consecutive b's	4 4	Section-I
2	a) Consider the CFG: S → aB | bA, A → a | aS | bAA, B → b | bS | aBB Find LMD and RMD for the string w= aaabbabbba b) Differentiate between DFA and NFA.	4 4	Section-I
3	a) Consider the following NFA and convert into DFA.  b) Consider the following NFA with ε and convert into DFA. 	4 4	Section-I
4	Convert the given NFA with epsilon to NFA without epsilon. 	8	Section-I
5	Describe the terms with examples. i. Regular Language ii. Regular Expression iii. Context free grammar iv. Derivation trees.	8	Section-I

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6	a) Draw NFA for the Regular expression: i) a(a+b)*ab ii) (0+1)*1(0+1) b) Draw DFA for the Regular expression: 10+(0+11)0*1	4 4	Section-I
7	What are the simplified forms in CFG? Explain each form in detail with example.	8	Section-I
8	Convert the following CFG into GNF. S → AA | 0 A → SS | 1	8	Section-I
9	a) Design a FA from given regular expression ab + (a + bb) a*b. b) Write the regular expression for the language accepting all the string which are starting with 1 and ending with 0, over ∑ = {0, 1}.	4 4	Section-I
10	a) What is Ambiguous grammar? Check the below grammar is ambiguous or not using the string "a (a) (a) a". A→AA A→(A) A→a b) Convert the following CFG into CNF. S → ASB A → aAS|a|ε B → SbS|A|bb	4 4	Section-I
11	Discuss the different phases of a compiler.	8	Section-II
12	a) Explain in detail about input buffering with example. b) Discuss about One-buffer scheme and Two buffer scheme with examples.	4 4	Section-II
13	Explain various operations on string with example.	8	Section-II
14	Explain about specification and Recognition of Tokens.	8	Section-II
15	Explain LEX program structure and Write a LEX program to recognize tokens. (Identifiers, keywords, relational operators and numbers).	8	Section-II
16	Explain in detail about design of lexical analyzer generator.	8	Section-II
17	a) Explain the roles of lexical analyser. b)Explain LEX program structure with example LEX program	4 4	Section-II

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18	Describe the terms. i) Symbol Table ii) Tokens iii) Lexemes iv) Patterns v) Parse Tree	8	Section-II
19	Differentiate between Compiler and Interpreter.	8	Section-II
20	Define the Phases of a compiler which indicating the inputs and outputs of each phase in translating the statement “x=y+z*40”.	8	Section-II
21	What is back tracking? Write recursive procedures for each non terminal by taking suitable example.	8	Section-III
22	Calculate FIRST and FOLLOW from the following grammar. S→ABCDE A→ a | Є B→ b | Є C→ c D→ d | Є E→ e | Є	8	Section-III
23	Construct LR parsing table for the following grammar. S → CC C → aC | d	8	Section-III
24	Construct SLR parsing table for the following grammar. S → AA A → aA | b.	8	Section-III
25	Calculate FIRST and FOLLOW from the following grammar. E -> TE’ E’ -> +T E’|Є T -> F T’ T’ -> *F T’ | Є F -> (E) | id	8	Section-III
26	Discuss about Bottom up parsing with an example.	8	Section-III

QUESTION BANK

27	What is meant by Parsing? Discuss about Top down parsing and Bottom up parsing with an example.	8	Section-III
28	a) Explain in detail about the Role of Parser in Compiler. b) Write a short note on error recovery with LR parsers. How is it different from LL parsers?	4 4	Section-III
29	Construct LL (1) parsing table for the following grammar. E->E+T |T T->T*F|F F->(E) |id	8	Section-III
30	Discuss about Shift reduce parsing with an example.	8	Section-III
31	What are different intermediate code forms? Explain with example.	8	Section-IV
32	Construct a Three-address code, quadruples, triples and Syntax tree for the following expression: a = b* - c + b * -c ;	8	Section-IV
33	Define SDT and write its applications.	8	Section-IV
34	What are different types of attributes? Explain with example.	8	Section-IV
35	Describe the construction of syntax tree. Construct a syntax tree for the expression: x*y-5+z	8	Section-IV
36	Explain the Variants of Syntax tree.	8	Section-IV
37	a) Distinguish between synthesized attributes and inherited attributes. b) Construct the syntax tree for the expression a-4+c*d.	4 4	Section-IV
38	Discuss different Three Address code types and implementations of Three Address statements.	8	Section-IV
39	a) Elaborate the role of intermediate code generator in compilation process. b) Explain in detail about Abstract syntax tree.	4 4	Section-IV
40	What are Boolean Expressions? And Write Syntax Directed Translation for the Boolean Expressions: **E → E OR E** **E → E AND E** **E → id1 relop id2** **E → NOT E**	8	Section-IV

QUESTION BANK

	E → (E) **E → FALSE** **E → TRUE**		
41	What is the role of code Optimizer in compiler? Explain various machine independent code optimization techniques.	8	Section-V
42	Explain in detail about various peephole optimizations.	8	Section-V
43	a) Explain the role of DAG in optimization with suitable example. b) Explain the Redundant sub expression elimination with an example.	4 4	Section-V
44	What is the purpose of loop optimization? Explain in detail loop optimization with example.	8	Section-V
45	Describe the terms with examples. i. Code motion ii. Copy propagation iii. Dead code Elimination iv. DAG v. Flow Graphs	8	Section-V
46	What is the purpose of optimization of building blocks? Explain different sources of optimization techniques with suitable examples.	8	Section-V
47	Explain how loop optimization can be done? How they are different from local optimizations.	8	Section-V
48	Explain the importance of register allocation with respect to optimization.	8	Section-V
49	a) Explain in detail about simple code generator. b) Discuss in detail the role of dead code elimination and strength reduction during code optimization.	4 4	Section-V
50	Explain in detail about basic blocks and flow graphs with an example.	8	Section-V